Answer Key - Physics - Graduate Level Assessment 25 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 1.0 diopter; B) 4.0 diopters; C) 2.0 diopters; D) 2.75 diopters; E) 3.5 diopters
   Note: Correct option: A - 1.0 diopter

2. **Answer:** C
   Options: A) The presence of moons around several planets; B) Planets orbit around the Sun in nearly circular orbits in a flattened disk.; C) the equal number of terrestrial and jovian planets; D) The existence of the Sun's magnetic field; E) The different chemical compositions of the planets
   Note: Correct option: C - the equal number of terrestrial and jovian planets

3. **Answer:** A
   Options: A) only when there are no charges outside the surface; B) only when the charges are in motion; C) only when the electric field is uniform; D) only when the enclosed charge is negative; E) only when the charge is at rest
   Note: Correct option: A - only when there are no charges outside the surface

4. **Answer:** A
   Options: A) remains the same; B) none of these; C) hotter; D) the temperature oscillates; E) it gets colder, but only under low pressure
   Note: Correct option: A - remains the same

5. **Answer:** D
   Options: A) -2/5 cm; B) -3/2 cm; C) -2/3 cm; D) -5/2 cm; E) -5 cm
   Note: Correct option: D - -5/2 cm

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'decreases nonlinearly'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 3 mm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $3.03 \times 10^-19 joule.Explainwhythisisthecorrectanswerandprovidesupportingevidence.Re$
    Note:$Long-formderivedfromMCQ;awardcreditforstatingthecorrectideaandgivingsupportingrea$

End of Answer Key